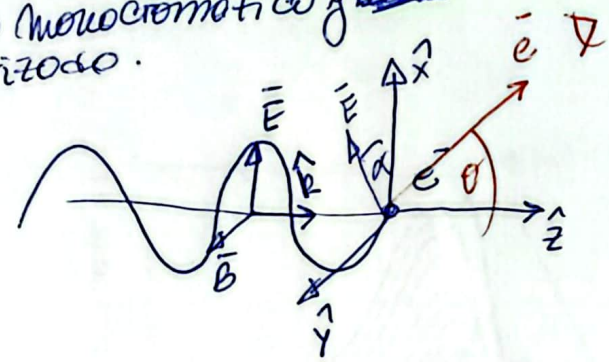
Thomson scattering cross section

onda monocromática y polarizada.



$\bar{e}$ ∇ e^- libre

. $\bar{e}$ dirección de propagación de la onda irradiada

$$m\ddot{\bar{x}} = mc\dot{\bar{\beta}} = -e\bar{E} - \underbrace{\frac{e}{c}\bar{v}\times\bar{B}}_{\text{despreciable si } v \ll c}$$

caso no relativista

$\bar{e}$ acelerado $\Rightarrow$ emite radiación

Potencia irradiada por unidad de ángulo sólido

$$\frac{dP}{d\Omega} = \frac{e^2}{4\pi c}\left|\bar{e}\times\dot{\bar{\beta}}\right|^2$$

fórmula de Larmor no relativista

$$\dot{\bar{\beta}} = -\frac{e}{mc}E\begin{pmatrix} c\alpha \\ s\alpha \\ 0 \end{pmatrix} \qquad \bar{e} = \begin{pmatrix} s\theta \\ 0 \\ c\theta \end{pmatrix}$$

$$\bar{e}\times\dot{\bar{\beta}} = -\frac{e}{mc}E\begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ s\theta & 0 & c\theta \\ c\alpha & s\alpha & 0 \end{vmatrix} = -eE\left(-s\alpha c\theta\,\hat{x} + \hat{y}\,c\theta c\alpha + s\theta s\alpha\,\hat{z}\right)$$

$$\frac{dP}{d\Omega} = \frac{e^4E^2}{4\pi c^3m^2}\Big(\underbrace{s^2\alpha c^2\theta + c^2\theta c^2\alpha}_{c^2\theta} + s^2\theta s^2\alpha\Big) = \frac{e^4E^2}{4\pi m^2c^3}\left(1 - s^2\theta c^2\alpha\right)$$

$$1 - s^2\theta + s^2\theta s^2\alpha = 1 - s^2\theta\left(1 - s^2\alpha\right) = 1 - s^2\theta c^2\alpha$$

Escaneado con CamScanner

$$\frac{dP}{d\Omega} = \overbrace{|\bar{S}|}^{\sim \phi_b \times N_t} \frac{d\sigma}{d\Omega}$$

tomo esto como definición de la sección eficaz $\longrightarrow$

$\frac{dR_i}{d\Omega}$ el flujo $\longrightarrow$ el vector de Poynting

$$\bar{S} = \frac{1}{\mu} \bar{E} \times \bar{B} = \bar{E} \times \bar{H}$$

$$\mu_0 = 4\pi \times 10^{-7} \ T\,m\,A^{-1}$$

plano electromagnético propagándose en el vacío $\bar{B} = \frac{1}{c} \hat{k} \times \vec{E}$

$$\Rightarrow \bar{S} = \frac{1}{c\mu_0} |E|^2 \hat{k}$$

$$c^2 = \frac{1}{\mu_0 \epsilon_0}$$

$$\frac{1}{c\mu_0} = \frac{\cancel{c}\epsilon_0}{\cancel{c}} = c\epsilon_0$$

$$\frac{e^4}{4\pi m^2 c^3} \cancel{E^2} \left(1 - \sin^2\theta \cos^2\alpha\right) = \overbrace{\boxed{\frac{1}{c\mu_0}}}^{c\epsilon_0} \cancel{E^2} \frac{d\sigma}{d\Omega}$$

$$\underbrace{\frac{e^4}{4\pi m^2 \epsilon_0 c^4}}_{r_0^2} \left(1 - \sin^2\theta \cos^2\alpha\right) = \frac{d\sigma}{d\Omega}$$

¡me falta algo en esta cte?

Escaneado con CamScanner

luz no polarizado

$$\left\langle \frac{d\sigma}{d\Omega} \right\rangle_\alpha \longrightarrow \frac{1}{2\pi}\int_0^{2\pi} 1 - s_\theta^2 c_\alpha^2 \, d\alpha = \left(2\pi - \pi s_\theta^2\right)\frac{1}{2\pi}$$

$$\int_0^{2\pi} c^2\alpha \, d\alpha = \int_0^{2\pi} \frac{c(2\alpha)+1}{2} \, d\alpha = \frac{1}{2}\left[\frac{s(2\alpha)}{2} + \alpha\right]_0^{2\pi} = \pi$$

$$C(2\alpha) = c^2\alpha - s^2\alpha = c^2\alpha - (1 - c^2\alpha) = 2c^2\alpha - 1$$

$$1 - \frac{s^2\theta}{2} = \frac{2 - s^2\theta}{2} = \frac{2 - (1 - c^2\theta)}{2} = \frac{1 + c^2\theta}{2}$$

$$\boxed{\frac{r_0^2}{2}\left(1 + c^2\theta\right) = \frac{d\sigma}{d\Omega}}$$

<u>sección eficaz total</u>

$$\sigma_T = \int_0^{4\pi} \frac{d\sigma}{d\Omega} \, d\Omega = 2\pi \int_{-1}^{1} \frac{r_0^2}{2}\left(1 + c^2\theta\right) dc\theta$$

Escaneado con CamScanner

$$2\pi \frac{r_0^2}{2} \left[c\theta + \frac{c_\theta^3}{3} \right|_{-1}^{1} = \pi r_0^2 \left(\frac{2}{3} + 2 \right) = \boxed{\frac{8}{3} \pi r_0^2}$$

Sección eficaz diferencial - espectro energético

uso la relación de dispersión (*)

$$\frac{d\sigma}{dE'_\gamma} = \frac{d\sigma}{d\Omega} \frac{d\Omega}{dE'_\gamma} = \frac{r_0^2}{2} \left(1 + c^2\theta\right) \times (-2\pi) \frac{mc^2}{E'^2_\gamma}$$

(*) $$1 + c^2\theta = \left(mc^2 \left\{ \frac{1}{E_\gamma} - \frac{1}{E'_\gamma} \right\} + 1 \right)^2 + 1$$

c ?

$$\frac{d\sigma}{dE'_\gamma} = (-1) 2\pi \frac{r_0^2}{2} \frac{mc^2}{E'^2_\gamma} \left[\left(mc^2 \left\{ \frac{1}{E_\gamma} - \frac{1}{E'_\gamma} \right\} + 1 \right)^2 + 1 \right]$$

Thomson.

Escaneado con CamScanner